University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

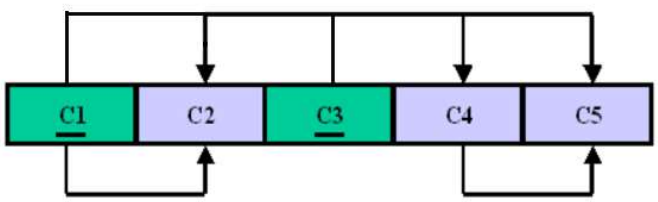
Assignment 4

Submission:

1. Use A4 size paper
2. Assignment should be neat and clean.
3. Submit the on portal (Scan in pdf form).
4. Each group has maximum 5 member.
5. Each member name and roll number mention clearly on assignment.
6. No late and retake submissions will be entertained.

Question 1: (10)

For the given dependency diagram identify the type for each of the given functional dependency. (i.e., Partial Functional Dependency, Full Functional Dependency, or Transitive Functional Dependency). Provide appropriate reasons for your choice.



Question 2: (30)

Put the relation below in normalized form (Till 3NF)

OID	O_Date	CID	C_Name	C_City	PID	P_Desc	P_Price	Qty
1006	10/24/21	2	Ali	LHR	7 5 4	Table, Desk, Chair	8000, 6250, 3000	1 1 5
1007	10/25/21	6	Ahmad	FSD	11 4	Dresser, Chair	5000, 3000	4 6

Where $OID = Order\ ID,\ O\_Date = Order\ Date,\ CID = Customer\ ID,$
$C\_Name = Customer\ Name,\ C\_City = Customer's\ City,\ PID = Product\ ID,$
$P\_Desc = Product\ Description,\ P\_Price = Product\ Price,$
$Qty = Quantity\ Purchased$

Note: 7, 5, 4 means three Product IDs. Similarly, 1, 1, 5 means three Quantities.

Functional Dependencies are:

$$OID -> O\_Date$$
$$CID -> C\_Name$$
$$PID -> P\_Desc$$
$$PID -> P\_Price$$
$$OID -> CID$$
$$CID -> C\_State$$
$$PID, OID -> Qty$$

Question 3: (30)

Order_ID	Date	Cust_ID	Cust_Name	State	City	ItemNo	ItemDescription	ItemPrice	Quantity
1001	5/6	A004	John	NY	New York	1234 6850 7230	Printer Keyboard Television	250 100 600	3 4 1
1002	3/6	A999	Tom	CA	San Francesco	7230	Television	600	2
1003	7/9	A301	David	CA	Los Angles	3434 1234	Scanner Printer	150 250	12 7
1004	3/10	A454	Rami	PA	Philadelphia	6850 3434 8876	Keyboard Scanner Speaker	100 150 70	1 1 2
1005	1/11	A301	David	CA	Los Angles	8876 9910	Speaker Microphone	70 80	1 1

- Explain what are the problems with the table above? Give some examples using the table data.
- Apply the normalization rules to make the above table in 3NF, define the functional dependencies and choose an appropriate primary key and re-design the table based on that.

Question 4: (20)

The following is a minimal cover for a relation R(ABCDEF):

$AB \rightarrow D$
$B \rightarrow C$
$CE \rightarrow B$
$D \rightarrow AF$

Identify the Normal form if there is need then apply normalization on given relation.

Question 5: (10)

Consider the following relation:

A	*B*	*C*
10	b1	c1
10	b2	c2
11	b4	c1
12	b3	c4
13	b1	c1
14	b3	c4

Given the data base state which of the following dependencies may hold in the above relation? If the dependency cannot hold, explain why by specifying the rows that cause the violation.

i. $A \rightarrow B$

ii. $B \rightarrow C$

iii. $C \rightarrow B$

iv. $B \rightarrow A$

v. $C \rightarrow A$